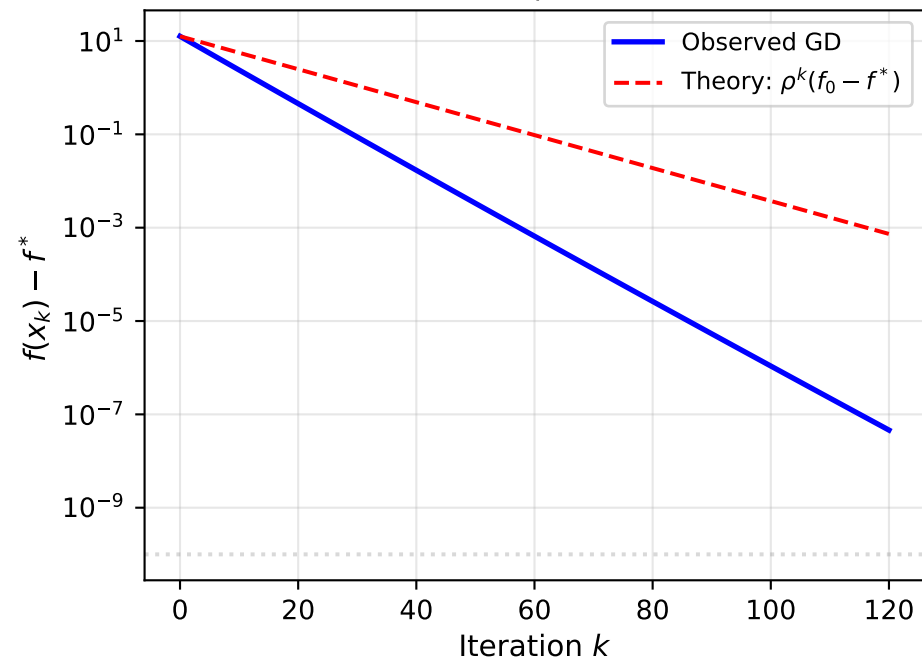
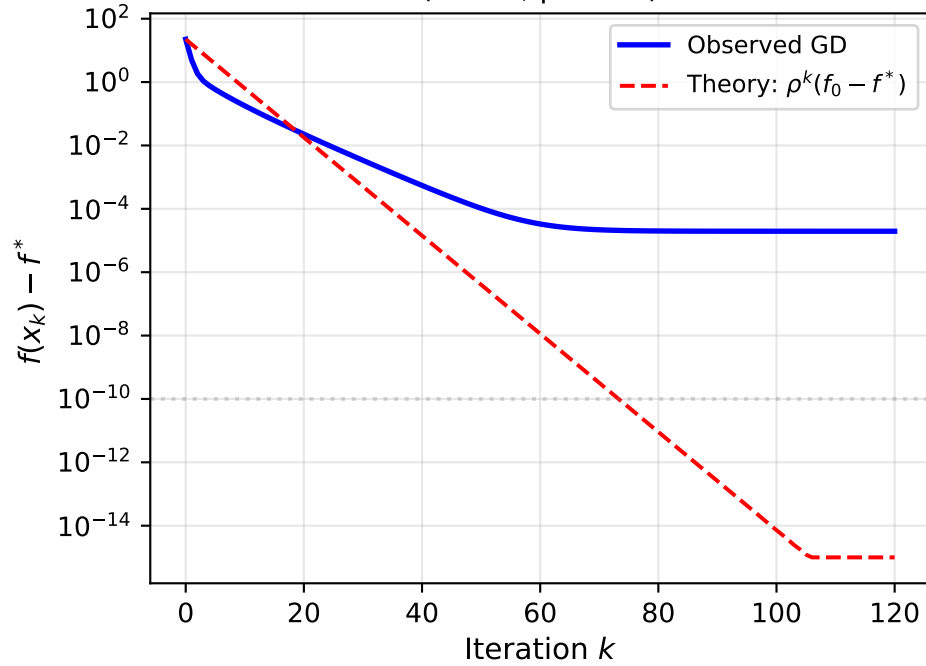


GD Convergence: Observed vs Theoretical Bound $f(x_k) - f^* \leq \rho^k(f(x_0) - f^*)$

Benchmark A
($\kappa \approx 1.12$, $\rho \approx 0.059$)



Benchmark B
($\kappa \approx 3.5$, $\rho \approx 0.55$)



Benchmark C
($\kappa \approx 2508$, $\rho \approx 0.9992$)

